HOME ADVANTAGE IN THE PREMIER LEAGUE

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INTRODUCTION

Football is like no other sport, competition, or event across the globe. It reaches billions around the world who play and watch the game, bypassing different races, religions, and cultures. And England's top division, the Premier League, is the top one percent of the one percent. Widely regarded as the best league in the world for competition and quality, it has been a major influence on the growth, evolution, and improvement of football. Recent figures suggest that each Premier League game has over 600 million viewers globally, with games between the biggest clubs reaching 700-750 million viewers (Moore, 2024). For context, the second-most-watched league is Spain's top division, LaLiga, with a reported 7.5 million viewers per game. And since its foundation in 1992, the Premier League has been the pride of football.

We are exploring the research question: Do home teams have an advantage over away teams in the Premier League? In this study, we used total shots taken, shots on target, number of fouls and corners in a match to answer this question. We also created new variables such as shot accuracy and "winner"(outcome). Predicting soccer matches is essential in the following fields. First, scientists use soccer predictions in dynamic networks. In dynamic networking, scientists consider soccer teams moving networks, players "nodes," and passes "edges." Through this, scientists predict how network density and connectivity lead to success, which has applications in epidemiology- the study of how viruses spread (Novillo et al,2024). Through this, scientists can make informed decisions in a viral outbreak and pandemic. Secondly, predicting soccer helps people who bet to make informed decisions and place winnable bets. This reduces financial losses. Lastly, soccer predictions enhance storytelling as the 600 million soccer enthusiasts and journalists use these statistics to make their opinions. In addition, soccer prediction is used by soccer analysts to identify areas that a team needs to improve, and to make good recommendations on recruitment.

DATA AND EXPLORATORY DATA ANALYSIS

This data was accessed by Ewan Gomer on Kaggle. Gomer obtained the dataset from the English Premier League data and cleaned it using Google Sheets. Kaggle, on the other hand, is an online public database that allows users to explore data, build machine learning models and participate in data science competitions.

The variables relevant to our research question are full-time home goals, full-time away goals, home team shots, away team shots, home team shots on target, away team shots on target, home corners and winner. The winner of the match is categorical and compares the total number of goals scored by both teams and assigns the winner based on who scored more; draws are at times considered as an away team win. Home team winners are assigned 1, and away team wins are assigned 0. Table one contains information about our variables of interest, including the mean and standard deviation of each statistic for both home and away teams. As is evident in our table, almost every statistic has a mean that is higher for the home team than the away team. The home team generally takes more shots, scores more goals, and kicks more corner kicks than the away team.

Table 1

Name	Description	Mean	SD
Full-time home goals	Number of goals scored by the home team at the end of the match	1.513	1.326545
Full-time away goals	Number of goals scored by the away team at the end of the match	1.305	1.258836
Home shot	Total shots taken by the home team, which includes shots on target, blocked shots and shots off target	13.85	5.661784
Home shot on target	Home shots that hit the target	4.679	2.600402
Away shots	Total shots taken by the away team, which includes shots on target, blocked shots and shots off target	11.74	5.34846
Away shots on target	Away shots that hit the target.	4.142	2.589978
Home corners	Corners by the home team	5.603	3.029755
Away Corners	Corners by the away team	4.821	2.735399
Winner	The winner of the match, it is categorical and	n/a	n/a

	compares the total number of goals scored by both teams and assigns the winner based on who scored more.		
Home Team Winner	A categorical variable that returns true if the home team won in the first half and false if the away team led.	n/a	n/a

Table 2

This table contains the total counts of home, draws and away wins expressed as percentages. As is shown here, the home team wins nearly 10% more than the away team, suggesting that the home team may have an advantage over the visitors.

Full Time Result:	Home Win	Away Win	Draw	Away Win + Draw
Count	163	129	88	217
Proportions	42.89%	33.95%	23.16%	57.1%

Table 3

Using the formula- Team shots on target/Team shots*100, we obtained the team accuracy variable.

Name	**Description**	**Mean**	**SD**	**Max**	**Min**
Away Team Accuracy	The percentage of the shots taken by the away team that hit the target	35.33	16.54	*112.5	0
Home Team Accuracy	The percentage of the shots taken by the home team that hit the target	34.25	15.08	88.89	0

* Data from ESPN and the Premier League contradicts this.
"ESPN Match Day Statistics"

Figure 1

Table three is a stacked bar plot of full-time and half-time winners. Home team wins are represented in the left column, away team wins in the right column. In each bar, grey represents the count of the home team leading at halftime, and blue represents the count of the away team leading at halftime. The plot shows that the home team usually wins about two-thirds of the matches they win in the first half. The away team, on the other hand, wins about 10 percent of the matches that the home team won in the first half.

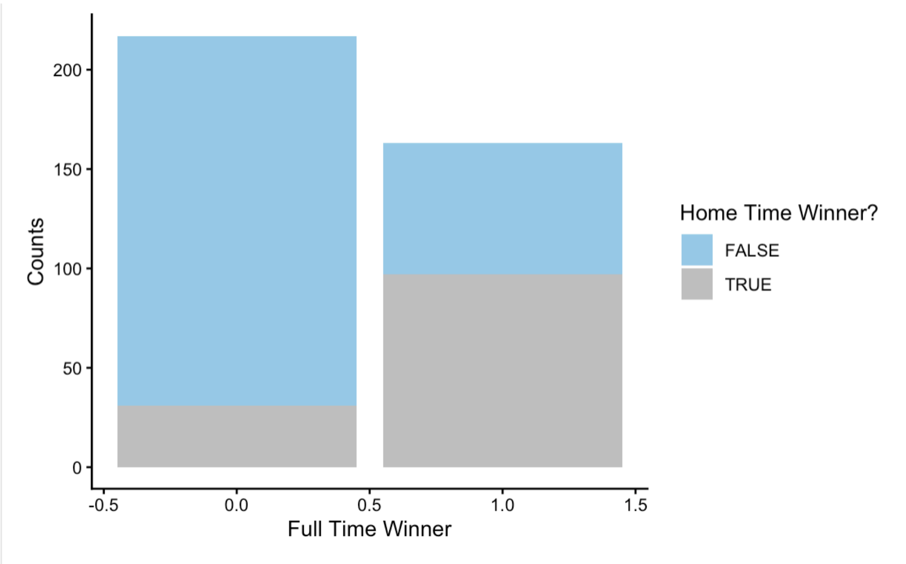


Table 5

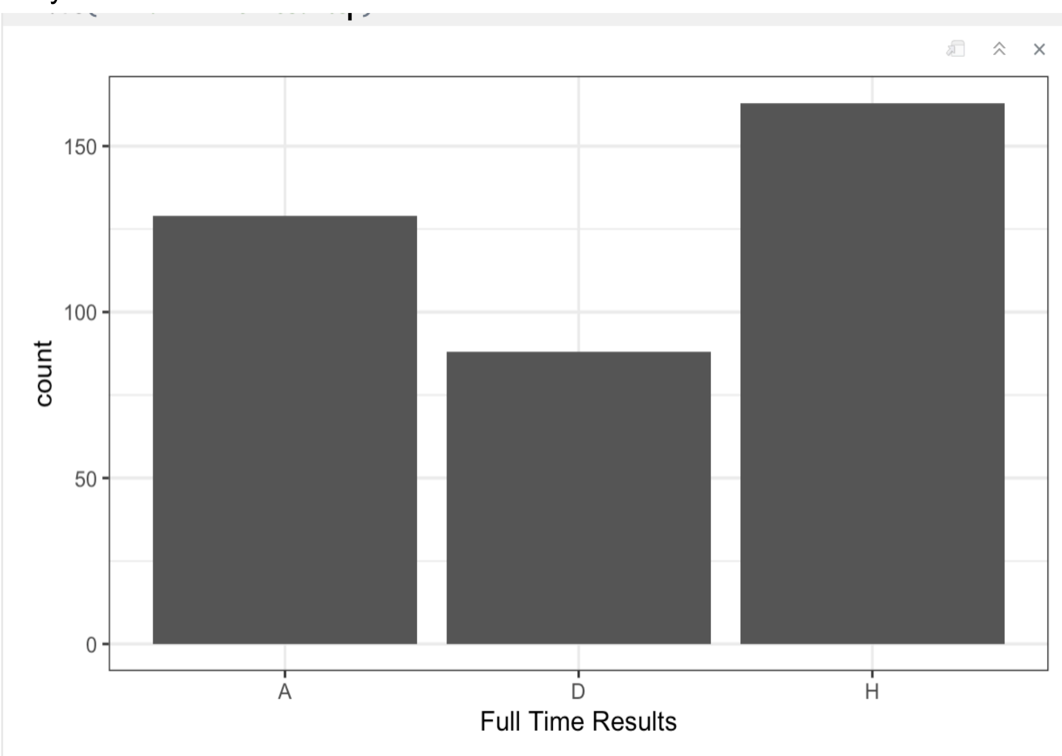
This table explores the change in a home team's winning probability based on the different variables. Although they are included, home/away shots and home/away fouls have no notable effect on the home team's winning chances. Additionally, the home/away corners have a counterintuitive per cent change, showing how the data may be flawed, along with the high P-Values for anything but home/away shots on target. Simply put, the two valid variables to analyse in this setting are home/away shots on target, which both also have a significant effect on the probability, whereas the foul and corner data can largely be discarded.

Home Team's Winning Probability Based on…		
Variable	Percent Change:	P Value:
Home Shot On Target	+69%	1.52e-11
Away Shot On Target	-41%	1.14e-08

Home Team’s Winning Probability Based on…		
Home Corner	-9% (counterintuitive)	0.0446
Away Corner	+17% (counterintuitive)	0.0100
Home Foul	-4% (slight negative)	0.3209
Away Foul	+0.1% (negligible)	0.9702
Home Shot	-2% (counterintuitive)	0.6696
Away Shot	-0.05% (negligible)	0.6833

Table 5

This figure shows the true proportion of the full-time results, with away wins, home wins and draws. This was the data we manipulated to create the winner variable by counting draws as away wins.



METHODS

Our logistic regression model is:

E[Winner|Home Team Full time Goals, Total Shots, Shots on Target, Home Team accuracy, Away team accuracy, Half time winner]=

β_0 + β_1(Home Team Full time Goals)+β_2(Total shots)+β_3(Shots on Target)+β_4(Home Team Accuracy)+β_5(Away Team Accuracy)+β_6(Half Time Winner)

"Winner" is our dependent variable. Home Team Goals is a quantitative variable, and it is the number of goals scored by the home team; it is our predictor of interest. Total shots and shots on target are confounding variables as they are the sum of the shots taken by both teams and therefore related to both of the teams. Home team and away team accuracy: how the teams are likely to hit the target. The away team's accuracy is higher than the home team's accuracy as the away team takes fewer shots. Half-time winner is a categorical variable. It is boolean in nature- It is true when the home team won the first half and false when the away team scored more or the half ended in a draw.

We added the halftime winner because of the nature of football/soccer. Some teams are known to turn their home ground into a "Fortress" and rarely lose at home. For example, Manchester United, one of the league's historic teams, last lost a game they were leading at halftime on May 7, 1984, against Ipswich Town. Given this fact, we decided to include the score at halftime. We also contemplated adding referees and teams as predictors. However, we excluded it because of the number of data points and a big difference between the games managed by the top referee and the bottom referees. Some referees were chosen to officiate as few as nine games, whereas others officiated nearly thirty games, and therefore, including these outliers may distort our results, as the outliers skew key summaries such as mean and standard deviation and hence reduce the predictive power of models. We selected a significance threshold of 0.05 for the hypothesis test and 0.5 for the confusion matrix.

Our null hypothesis is that while adjusting for shots and accuracy, there is no relationship between goals scored by the home team and the outcome of the match.

H_0: $\beta_1=\beta_2=\beta_3=\beta_4=\beta_5=\beta_6=0$

Our alternative hypothesis is: There is a relationship between the outcome of the match and goals scored by the home team(Adjusting for shots and accuracy)

H_A: At least one of $\beta_1,\beta_2,\beta_3,\beta_4,\beta_5,\beta_6\neq0$

To evaluate our model, we used the confusion matrix, nested model and an F-test.

RESULTS

Coefficients	**Estimates**	**Std. Error**	**P value**
β_0(Intercept)	0.345	4.794	1.644
β_1(Home Team Full time Goals)	10.361	1.318	1.000
β_2(Total shots)	1.001	1.065	2.676
β_3(Shots on Target)	0.728	1.194	1.077
β_4(Home Team Accuracy)	1.028	1.025	1.295
β_5(Away Team Accuracy)	0.962	1.0194	1.050
β_6(Half Time Winner)	2.390	1.403	1.010

*exponentiated odds

On average, while holding the other variables constant, we predict that the odds of a team winning are 0.343. For the home goals, our model predicts that the odds of the home team winning are multiplied by 10.361 for every goal they score. Given a confidence interval of 95%, while holding all the other variables constant, we are sure that the odds of winning are affected by home full-time goals by a multiplicative value that is between 1.834 and 2.92.

Using two nested models, we concluded that we have significant evidence that home goals have a relationship with the outcome of the match, given the p-value is 2.2 × 10^-16, which is significantly below our threshold of 0.05; therefore, we reject the null hypothesis.

Confusion Matrix-0.5 Threshold

Outcome	Predicted 0	Predicted 1	Total
0	188	29	217
1	33	130	163
Total	221	159	380

Based on the data from the table, we found that the accuracy of our model was at 88.33%, the sensitivity was at 79.75%, and the specificity was 86.64%. This shows that our model correctly predicted the outcome of a match 88.33 times out of 100. Interestingly, the model is better at predicting away wins than home wins.

CONCLUSION

Exploring the data from the Premier League 2020-2021 season, we can conclude that the home team does have an advantage over away teams as they boast a win rate of 42.89% compared to 33.95% win rate of away sides. Home teams also win about 90% of the matches they lead at halftime. As per the model, a home team's goals significantly affect the outcome of the match. However, there are only a handful of variables that contribute significantly to this outcome. Our data does have some limitations in that we do not address some factors that could be important contributors to the home team advantage. A 2008 study by Richard Pollard discusses some potential factors, including crowd effects, travel effects on away players, familiarity (with the city/stadium), and psychological factors. These are confounding variables that are hard to empirically measure, are not included in our model, and can have an effect on our data that we cannot account for. We kept our data as practical as possible, and did not include any variables that are multicollinear or redundant. Some variables were reused or modified (e.g., shot accuracy) in order to get a more expansive understanding of home team advantage. The consequences of these shortcomings result in an incomplete outcome that should be tested by others and/or with different methods. As for ethical issues, Betting, an application of soccer prediction that we discussed in the introduction, can be addictive, with people losing thousands of dollars as they try to beat the odds in the hope of winning.

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